Problem 26.6

An annual annuity-immediate pays $100 at the end of the first year. Each subsequent payment is 5% greater than the preceding payment. The last payment is at the end of the 20th year. Calculate the accumulated value at:

(a) an annual effective interest rate of 4%;

(b) an annual effective interest rate of 5%.

Solution

The payments are

$$100,\ 100(1.05),\ 100(1.05)^2, \ldots, 100(1.05)^{19}.$$

The accumulated value at time 20 is

$$AV = 100(1+i)^{19} + 100(1.05)(1+i)^{18} + \cdots + 100(1.05)^{19}.$$

This is a geometric sum, so when $i \neq 0.05$,

$$AV = 100\left(\frac{(1+i)^{20} - (1.05)^{20}}{i - 0.05}\right).$$

Part (a).

For $i = 0.04$,

$$AV = 100\left(\frac{(1.04)^{20} - (1.05)^{20}}{0.04 - 0.05}\right).$$

Thus,

$$AV = 4621.745621.$$

Therefore,

$$\boxed{4621.75}.$$

Part (b).

For $i = 0.05$, the interest rate is equal to the payment growth rate. Therefore, each payment accumulates to the same value at time 20:

$$100(1.05)^{19}.$$

Since there are 20 payments,

$$AV = 20(100)(1.05)^{19}.$$

Thus,

$$AV = 5053.900391.$$

Therefore,

$$\boxed{5053.90}.$$